MNXB11

Final Project

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1 Introduction

2 Numerical Methods

2.1 Cleaning of Raw Data

The data was cleaned in bash. The bash program takes the .txt data in and splits the columns into year, month, day, hour, minute, second, temperature and grading. This is done, so that all data is clearly identified in later stages of the analysis. All comments are taken are not taken into account. Finally the bash program creates a `cleaned_filename.csv` in the datasets directory with measurements in the format `YYYY-MM-DD HH:MM:SS Temperature Y/G` .

2.2 Creating a Root Tree

The file `src/run_weather.cxx` creates a root tree by accessing the specified csv file, separating the year, month, day, hour, minute, second, temperature and grading and writing each to a branch of the tree called `MNXB11_Temperature` stored in a root-file called `weather_data.root`. If there is data that is not correctly formatted, which means that numbers do not build valid dates and times, then the data is not written to the tree. Measurements before 1500 and after 2026 are also rejected.

3 Analytical Methods

3.1 Yearly Average

The yearly average is found by taking the mean over all measurement points during one year. If the year has no measurements it is excluded from the measurements and not considered in further calculations. The same is done for years that have an unequal distribution of measurement points during the year, which is evaluated by looking at the number of measurement points per month and allowing measurements for which the difference between the maximum and minimum measurement points per month and year is smaller than 1/3 of the maximum points.
The mean over the whole measurement period is calculated with all years with valid mean temperatures. The running average is calculated over a period of ± 10 years from the center year. If there were years with excluded or no measurements the running average was still computed but no data was included from those years.

3.2 Daily Average

3.3 Extreme temperatures

To extract yearly extreme temperatures from the datasets, the program first makes a guess that every year the max temp was -9999 ^oC and min was 9999 ^oC. It then goes through the entire data set entry by entry, comparing if the stored min/max for the year is higher/lower than the current entry. If the entry is found to be a more extreme temperature, the old value is thrown out and replaced. The month the new extreme value was recorded in is also saved for the plotting. The same is also done but for each month of the year to find the monthly extremes.

4 Result and Analysis

4.1 Yearly Average

The yearly average from Lund is portrayed in figure 1. It shows that the mean temperature over the whole measurement period was $T = 8°C$ and is shown as the dashed line. The running average over 21 years is portrayed as the black line. Some histogram bins are not filled and instead show a gray background, because they did not fulfill the analysis requirements as described above.

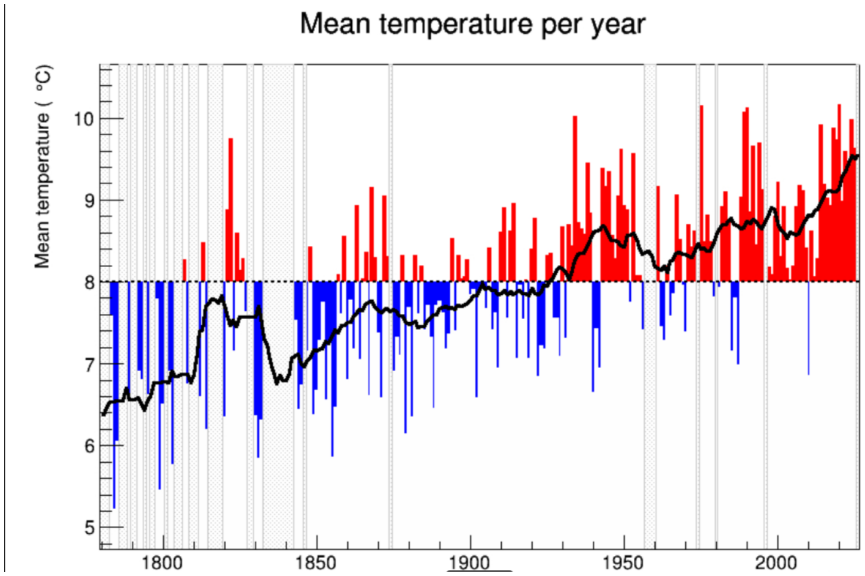


Figure 1: The mean annual temperature in Lund with running average and overall mean temperature.

The running average increases over the 200 years by around 3 °C. In the beginning of the measurement period most average temperatures lie below the average. Then the average temperature increases and today most of the temperatures lie above the longtime mean.

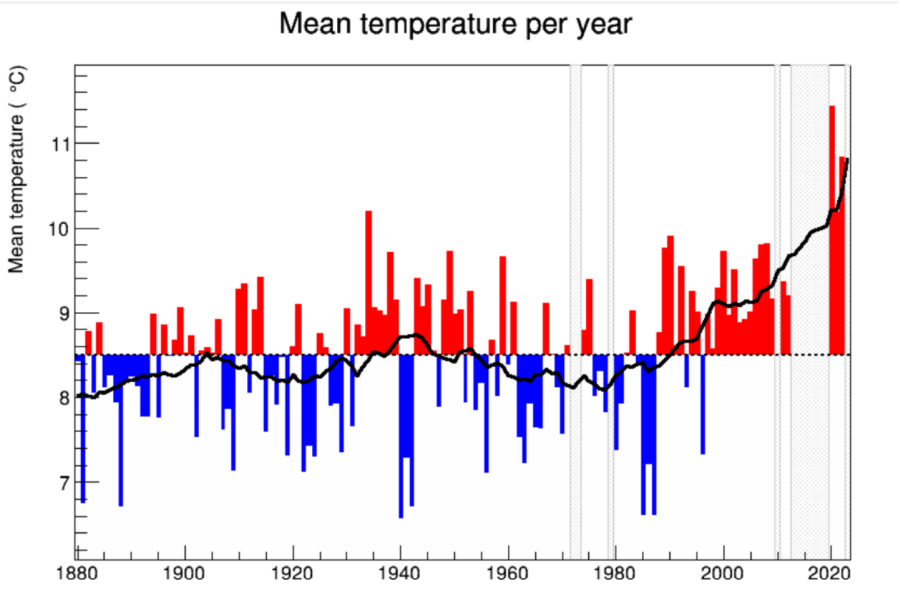


Figure 2: The mean annual temperature in Falsterbo with running average and overall mean temperature.

4.2 Daily Average

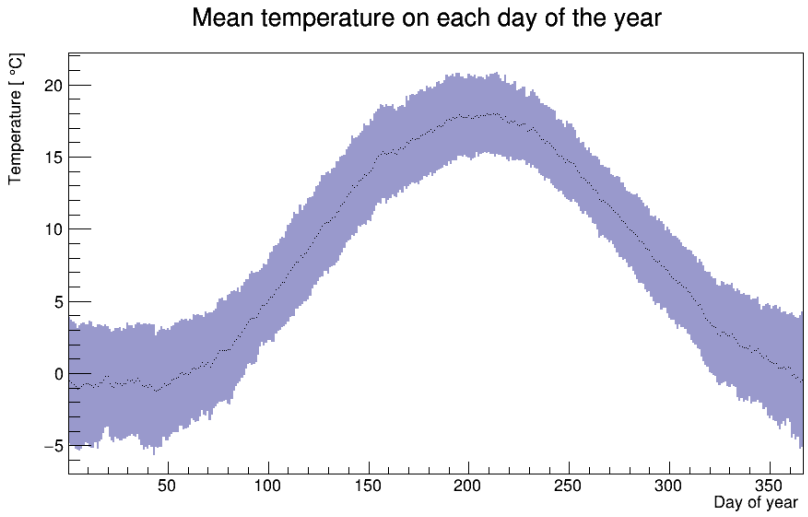


Figure 3: The mean daily temperature in Lund on each day of the year, and its standard deviation.

4.3 Extreme temperatures

5 Discussion

5.1 Yearly Average

The results for Lund and Falsterbo agree with the findings of SMHI that compared with 1890 the temperatures have been raised by 1.9°C on average [1].Moreover, agree both measurements with each other showing a consistent picture of the climate developments

in Skåne. Both show a big increase of the running average in the last 40 years. Both plots show the same behavior considering that the mean temperature in Falsterbo is 0.5°C higher than in Lund, because the data starts 100 years later.

5.2 Daily Average

5.3 Extreme temperatures

6 Conclusion

7 References

[1] Semjon Schimanke et al. *Observerad klimatförändring i Sverige 1860–2021*. Tech. rep. 69. SMHI Report Klimatologi 69. Sveriges meteorologiska och hydrologiska institut (SMHI), 2022. URL: https://www.smhi.se/publikationer-fran-smhi/sok-publikationer/2022-11-20-observerad-klimatforandring-i-sverige-1860-2021.

8 Appendix